General Linear Models Excercise

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```
veteran <- survival::veteran
```

The survival function allows RMarkdown to track time to event and event indicators for data in the table. In this case, the event is survival or death.

note:status is 0 if patient survives, 1 if they died

prior: prior therapy 0 if no, 10 if yes

```
veteran_clean<- veteran %>%
  mutate(
    trt = factor(trt,
                 levels = c(1,2),
                 labels = c("Standard", "Test")),
    prior = factor(prior,
                   levels = c(0, 10),
                   labels = c("No", "Yes")),
    status = factor(status,
                    levels = c(0,1),
                    labels = c("Alive", "Dead")),
    celltype = factor(celltype,
                      levels = c("squamous", "smallcell", "adeno", "large"),
                      labels = c("Squamous", "Small Cell", "Adenocarcinoma", "Large Cell")),
    survival1year = if_else(time >= 365, 0, 1)
   )

vctb<-as.tibble(veteran_clean)
```

```
## Warning: `as.tibble()` was deprecated in tibble 2.0.0.
## i Please use `as_tibble()` instead.
## i The signature and semantics have changed, see `?as_tibble`.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
vctb
```

```
## # A tibble: 137 x 9
##    trt      celltype  time status karno diagtime   age prior survival1year
##    <fct>    <fct>    <dbl> <fct>  <dbl>    <dbl> <dbl> <fct>          <dbl>
##  1 Standard Squamous    72 Dead      60        7    69 No                 1
##  2 Standard Squamous   411 Dead      70        5    64 Yes                0
##  3 Standard Squamous   228 Dead      60        3    38 No                 1
##  4 Standard Squamous   126 Dead      60        9    63 Yes                1
##  5 Standard Squamous   118 Dead      70       11    65 Yes                1
##  6 Standard Squamous    10 Dead      20        5    49 No                 1
```

```
##  7 Standard Squamous    82 Dead      40       10    69 Yes              1
##  8 Standard Squamous   110 Dead      80       29    68 No               1
##  9 Standard Squamous   314 Dead      50       18    43 No               1
## 10 Standard Squamous   100 Alive     70        6    70 No               1
## # i 127 more rows
```

The mutate function turns the trt, prior, and status values into named factors.

I also created a new variable, survival1year, which tracks survival of patients 1 year after treatment.

General Linear Models

```
model_1<- glm(survival1year ~ trt, family = binomial, data = vctb)

model_2<- glm(survival1year ~ trt + age + karno, family = binomial, data = vctb)

model_3<- glm(survival1year ~ trt + celltype + karno + diagtime + age + prior, family = binomial, data =

model_4 <- glm(survival1year ~ trt*celltype + karno + age + diagtime + prior, family = binomial, data =
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
model_5<- glm(survival1year~ trt + age + karno + diagtime + prior, family = binomial, data = vctb)

summary(model_1)
```

```
##
## Call:
## glm(formula = survival1year ~ trt, family = binomial, data = vctb)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   2.7881     0.5151   5.412 6.22e-08 ***
## trtTest      -0.4527     0.6695  -0.676    0.499
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71.600  on 136  degrees of freedom
## Residual deviance: 71.133  on 135  degrees of freedom
## AIC: 75.133
##
## Number of Fisher Scoring iterations: 5
```

```
summary(model_2)
```

```
##
## Call:
## glm(formula = survival1year ~ trt + age + karno, family = binomial,
##     data = vctb)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  5.399770   2.547046   2.120   0.0340 *
## trtTest     -0.460128   0.688449  -0.668   0.5039
```

```
## age          0.005174   0.033784   0.153   0.8783
## karno       -0.044535   0.021406  -2.080   0.0375 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71.60  on 136  degrees of freedom
## Residual deviance: 65.59  on 133  degrees of freedom
## AIC: 73.59
##
## Number of Fisher Scoring iterations: 6
```

```
summary(model_3)
```

```
##
## Call:
## glm(formula = survival1year ~ trt + celltype + karno + diagtime +
##     age + prior, family = binomial, data = vctb)
##
## Coefficients:
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)               2.37423    2.85516   0.832   0.4057
## trtTest                  -0.34022    0.76741  -0.443   0.6575
## celltypeSmall Cell        1.46617    0.96812   1.514   0.1299
## celltypeAdenocarcinoma   18.34444 1925.13788   0.010   0.9924
## celltypeLarge Cell        1.06592    0.93142   1.144   0.2525
## karno                    -0.03968    0.02363  -1.679   0.0931 .
## diagtime                  0.20053    0.12736   1.575   0.1154
## age                       0.01682    0.03814   0.441   0.6592
## priorYes                 -0.95532    1.05527  -0.905   0.3653
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71.600  on 136  degrees of freedom
## Residual deviance: 53.186  on 128  degrees of freedom
## AIC: 71.186
##
## Number of Fisher Scoring iterations: 18
```

```
summary(model_4)
```

```
##
## Call:
## glm(formula = survival1year ~ trt * celltype + karno + age +
##     diagtime + prior, family = binomial, data = vctb)
##
## Coefficients:
##                                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)                     3.607e+00  3.129e+00   1.153    0.249
## trtTest                        -1.250e+00  1.271e+00  -0.984    0.325
## celltypeSmall Cell              2.195e-01  1.356e+00   0.162    0.871
## celltypeAdenocarcinoma          1.836e+01  5.571e+03   0.003    0.997
```

```
## celltypeLarge Cell              5.075e-01  1.603e+00   0.317    0.752
## karno                          -3.866e-02  2.456e-02  -1.574    0.115
## age                             6.249e-03  4.069e-02   0.154    0.878
## diagtime                        1.940e-01  1.209e-01   1.605    0.109
## priorYes                       -9.680e-01  1.043e+00  -0.928    0.353
## trtTest:celltypeSmall Cell      1.892e+01  3.725e+03   0.005    0.996
## trtTest:celltypeAdenocarcinoma  1.322e+00  6.810e+03   0.000    1.000
## trtTest:celltypeLarge Cell      6.505e-01  2.079e+00   0.313    0.754
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71.600  on 136  degrees of freedom
## Residual deviance: 50.233  on 125  degrees of freedom
## AIC: 74.233
##
## Number of Fisher Scoring iterations: 19
```

```
summary(model_5)
```

```
##
## Call:
## glm(formula = survival1year ~ trt + age + karno + diagtime +
##     prior, family = binomial, data = vctb)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  4.305759   2.681411   1.606   0.1083
## trtTest     -0.452577   0.704947  -0.642   0.5209
## age          0.009844   0.034955   0.282   0.7782
## karno       -0.041121   0.021229  -1.937   0.0528 .
## diagtime     0.148322   0.106305   1.395   0.1629
## priorYes    -1.110223   0.935687  -1.187   0.2354
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71.600  on 136  degrees of freedom
## Residual deviance: 62.558  on 131  degrees of freedom
## AIC: 74.558
##
## Number of Fisher Scoring iterations: 7
```

I picked these models in order of increasing complexity.

- Model 1 predicts survival at one year based on only treatment type.
- Model 2 predicts survival at one year based on treatment type, age, and Karnofsky performance score.
- Model 3 predicts survival at one year based on all of the data from all of the variables.
- Model 4 predicts survival at one year based on all of the data from all of the variables given interaction between treatment and cancer cell type.
  - I decided to include interactions within Model 4, as the treatment type is always going to be dependent on the type of cancerous cells, in order to increase efficacy.
- Model 5 predicts survival at one year based on treatment, Karnofsky performance score, age, time of

diagnosis, and prior therapy.

```
AIC(model_1, model_2, model_3, model_4, model_5)
```

```
##         df      AIC
## model_1  2 75.13327
## model_2  4 73.59011
## model_3  9 71.18592
## model_4 12 74.23333
## model_5  6 74.55840
```

```
anova(model_1, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: survival1year
##
## Terms added sequentially (first to last)
##
##
##      Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL                   136     71.600
## trt   1  0.46628       135     71.133   0.4947
```

```
anova(model_2, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: survival1year
##
## Terms added sequentially (first to last)
##
##
##       Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL                    136     71.600
## trt    1   0.4663       135     71.133  0.49470
## age    1   0.1127       134     71.021  0.73709
## karno  1   5.4305       133     65.590  0.01979 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_3, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: survival1year
##
## Terms added sequentially (first to last)
##
##
##          Df Deviance Resid. Df Resid. Dev Pr(>Chi)
```

```
## NULL                       136     71.600
## trt       1   0.4663       135     71.133  0.49470
## celltype  3   8.5873       132     62.546  0.03531 *
## karno     1   4.7565       131     57.789  0.02919 *
## diagtime  1   3.5040       130     54.285  0.06122 .
## age       1   0.2701       129     54.015  0.60329
## prior     1   0.8294       128     53.186  0.36245
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_4, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: survival1year
##
## Terms added sequentially (first to last)
##
##
##              Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL                         136     71.600
## trt           1   0.4663       135     71.133  0.49470
## celltype      3   8.5873       132     62.546  0.03531 *
## karno         1   4.7565       131     57.789  0.02919 *
## age           1   0.1009       130     57.688  0.75075
## diagtime      1   3.6732       129     54.015  0.05529 .
## prior         1   0.8294       128     53.186  0.36245
## trt:celltype  3   2.9526       125     50.233  0.39899
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_5, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: survival1year
##
## Terms added sequentially (first to last)
##
##
##          Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL                       136     71.600
## trt       1   0.4663       135     71.133  0.49470
## age       1   0.1127       134     71.021  0.73709
## karno     1   5.4305       133     65.590  0.01979 *
## diagtime  1   1.6703       132     63.920  0.19621
## prior     1   1.3614       131     62.558  0.24330
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Several logistic regression models were fit to predict what caused a patient to survive or die within one year of treatment.

The treatment only mode showed almost no evidence that chemotherapy treatment changed odds of dying within one year. The Chi-squared test showed no significant evidence, and the AIC was very high indicating a bad model fit for the data.

I think that the best model was Model 3, as it had multiple variables with evidence of significance in the Chi-squared anova model and the strongest AIC score of roughly 71.186.

- It appears that the Karnofsky scores were an important covariate, as patients with higher Karnofsky scores had lower probability of death within one year of treatment. Cell Type also seemed to be an important covariate because patient survival changed based on the cancer type.
- Diagnosis time also seems to be significant, although not as significant as the other two variables.

It is important to note that AIC does not mean anything diagnostically on its own and must be used in conjunction with other tests in order to make sense.

```
summary_model3 <- summary(model_3)
oddsratios<- exp(coef(model_3))
conf_intervals<- exp(confint.default(model_3))

results1<- data.frame(
  Variable = names(oddsratios),
  OddsRatio = oddsratios,
  CI_Lower = conf_intervals[ ,1],
  CI_Upper = conf_intervals[ ,2],
  P_value = summary_model3$coefficients[,4]

)

results1
```

```
##                                     Variable    OddsRatio   CI_Lower
## (Intercept)                      (Intercept) 1.074272e+01 0.03988410
## trtTest                             trtTest 7.116157e-01 0.15813502
## celltypeSmall Cell         celltypeSmall Cell 4.332609e+00 0.64965145
## celltypeAdenocarcinoma celltypeAdenocarcinoma 9.265899e+07 0.00000000
## celltypeLarge Cell         celltypeLarge Cell 2.903516e+00 0.46784474
## karno                                 karno 9.610936e-01 0.91759572
## diagtime                           diagtime 1.222047e+00 0.95209591
## age                                     age 1.016961e+00 0.94371937
## priorYes                           priorYes 3.846878e-01 0.04862534
##                            CI_Upper   P_value
## (Intercept)            2893.535534 0.4056586
## trtTest                   3.202308 0.6575243
## celltypeSmall Cell       28.894730 0.1299134
## celltypeAdenocarcinoma         Inf 0.9923972
## celltypeLarge Cell       18.019664 0.2524532
## karno                     1.006653 0.0930867
## diagtime                  1.568538 0.1153676
## age                       1.095887 0.6591992
## priorYes                  3.043366 0.3653120
```

It is also important to note that the reference cancer type for cell type is squamous, represented as 1. This means that celltypeadeno is a comparison between the squamous and adeno cell type survival rates.

- This could mean that the reason the upper level of the CI for celltypeadeno is infinity is due to the fact that there is not enough data to reliably determine the effects of the adenocarcinoma cells on patient survival with squamous cells as a reference.

Seeing that model 4 had more interaction but performed more poorly than model 3, I created a new model to see whether the interaction between treatment and cell type was significant.

Null: The model interaction between treatment and cell type is not significant. Alternative: The model interaction between treatment and cell type is significant.

```
model_int<- glm(survival1year ~ trt*celltype, family = binomial, data = vctb)

model_reg<- glm(survival1year ~ trt + celltype, family = binomial, data = vctb)

summary(model_int)
```

```
##
## Call:
## glm(formula = survival1year ~ trt * celltype, family = binomial,
##     data = vctb)
##
## Coefficients:
##                                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)                     2.639e+00  1.035e+00   2.550   0.0108 *
## trtTest                        -1.540e+00  1.157e+00  -1.332   0.1830
## celltypeSmall Cell              1.402e-14  1.268e+00   0.000   1.0000
## celltypeAdenocarcinoma          1.693e+01  3.585e+03   0.005   0.9962
## celltypeLarge Cell              1.430e-14  1.464e+00   0.000   1.0000
## trtTest:celltypeSmall Cell      1.847e+01  2.535e+03   0.007   0.9942
## trtTest:celltypeAdenocarcinoma  1.540e+00  4.390e+03   0.000   0.9997
## trtTest:celltypeLarge Cell      1.299e+00  1.871e+00   0.694   0.4874
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71.600  on 136  degrees of freedom
## Residual deviance: 58.769  on 129  degrees of freedom
## AIC: 74.769
##
## Number of Fisher Scoring iterations: 18
```

```
summary(model_reg)
```

```
##
## Call:
## glm(formula = survival1year ~ trt + celltype, family = binomial,
##     data = vctb)
##
## Coefficients:
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)               1.8439     0.6335   2.910  0.00361 **
## trtTest                  -0.4422     0.6949  -0.636  0.52449
## celltypeSmall Cell        1.4790     0.8596   1.721  0.08534 .
## celltypeAdenocarcinoma   18.0315  2063.3764   0.009  0.99303
## celltypeLarge Cell        0.8991     0.8657   1.039  0.29901
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
```

```
##
##     Null deviance: 71.600  on 136  degrees of freedom
## Residual deviance: 62.546  on 132  degrees of freedom
## AIC: 72.546
##
## Number of Fisher Scoring iterations: 18
```

```
anova(model_reg, model_int, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: survival1year ~ trt + celltype
## Model 2: survival1year ~ trt * celltype
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1       132     62.546
## 2       129     58.769  3   3.7768   0.2866
```

A P-value of 0.288 is insignificant as it is much larger than standard P-value of 0.05. therefore, the model interaction between the cell type and treatment variables is not significant.

Based on the above tests, I conclude that model 3 is the best model at depicting the relationship between different cancer cell types and treatments and patient survival past 1 year. Karnofsky scores, cell type, and diagnosis time all seemed to be important covariates within the data. However, the interaction between cell types and treatment types was proven to be insignificant data.

Hazard tests

Some Proportional Hazards Models

A. Survival Rates at 90 days and 1 year

```
str(vctb[, c("time", "status", "trt")])
```

```
## tibble [137 x 3] (S3: tbl_df/tbl/data.frame)
##  $ time  : num [1:137] 72 411 228 126 118 10 82 110 314 100 ...
##  $ status: Factor w/ 2 levels "Alive","Dead": 2 2 2 2 2 2 2 2 2 1 ...
##  $ trt   : Factor w/ 2 levels "Standard","Test": 1 1 1 1 1 1 1 1 1 1 ...
```

```
table(vctb$status, useNA = "ifany")
```

```
##
## Alive  Dead
##     9   128
```

```
table(vctb$trt, useNA = "ifany")
```

```
##
## Standard     Test
##       69       68
```

```
summary(vctb$time)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1.0    25.0    80.0   121.6   144.0   999.0
```

```
vctb$time_num <- as.numeric(as.character(vctb$time))
vctb$status_num <- as.numeric(as.character(vctb$status))
```

```
## Warning: NAs introduced by coercion
```

```r
km_trt <- survfit(Surv(time, status) ~ trt, data = vctb)
summary(km_trt, times = c(90, 365))
```

```
## Call: survfit(formula = Surv(time, status) ~ trt, data = vctb)
##
##                 trt=Standard
##  time n.risk n.event Pr((s0)) Pr(Dead)
##    90     37      31   0.5467    0.453
##   365      4      29   0.0708    0.929
##
##                 trt=Test
##  time n.risk n.event Pr((s0)) Pr(Dead)
##    90     25      42     0.38     0.62
##   365      6      16     0.11     0.89
```

Based on the Kaplan Meier test, about 54% of people survive past 90 days with standard treatment, in comparison to 38% of surviving patients receiving chemotherapy treatment. At one year,7% of the standard treatment group are survivors, while 11% of the chemotherapy group survived.

Therefore, standard treatment has a higher short-term chance of survival for cancer patients than chemotherapy, but a slightly lower survival rate at 1 year of treatment. Chemotherapy isn't necessarily the better option when it comes to cancer survival, as it seems to simply have a different survival curve than standard treatment. It's also important to note that Kaplan Meier tests become inaccurate once the data collection period ends; they cannot predict future trends and therefore the data drops to zero, which is not accurate to real life. There is also no test of significance in Kaplan Meier testing, and more statistical analysis must be done to truly determine the difference in survival rates per treatment.

B. Restricted mean survival time at 90 days and 1 year

```r
vctb <-
  veteran_clean%>%
  transmute(
    time_int = time
    status_int = as.integer(status) - 1L,
    trt_int = as.integer(trt) - 1L)
#turning status and treatment numerics/labels into integers

table(vctb$status_int, useNA = "ifany")
table(vctb$trt_int, useNA = "ifany")

rmst_90 <- rmst2(
  time = vctb$time,
  status = vctb$status_rmst,
  arm = vctb$trt_rmst,
  tau = 90
)

rmst_365 <- rmst2(
  time = vctb$time,
  status = vctb$status_rmst,
  arm = vctb$trt_rmst,
  tau = 365
)


rmst_90
```

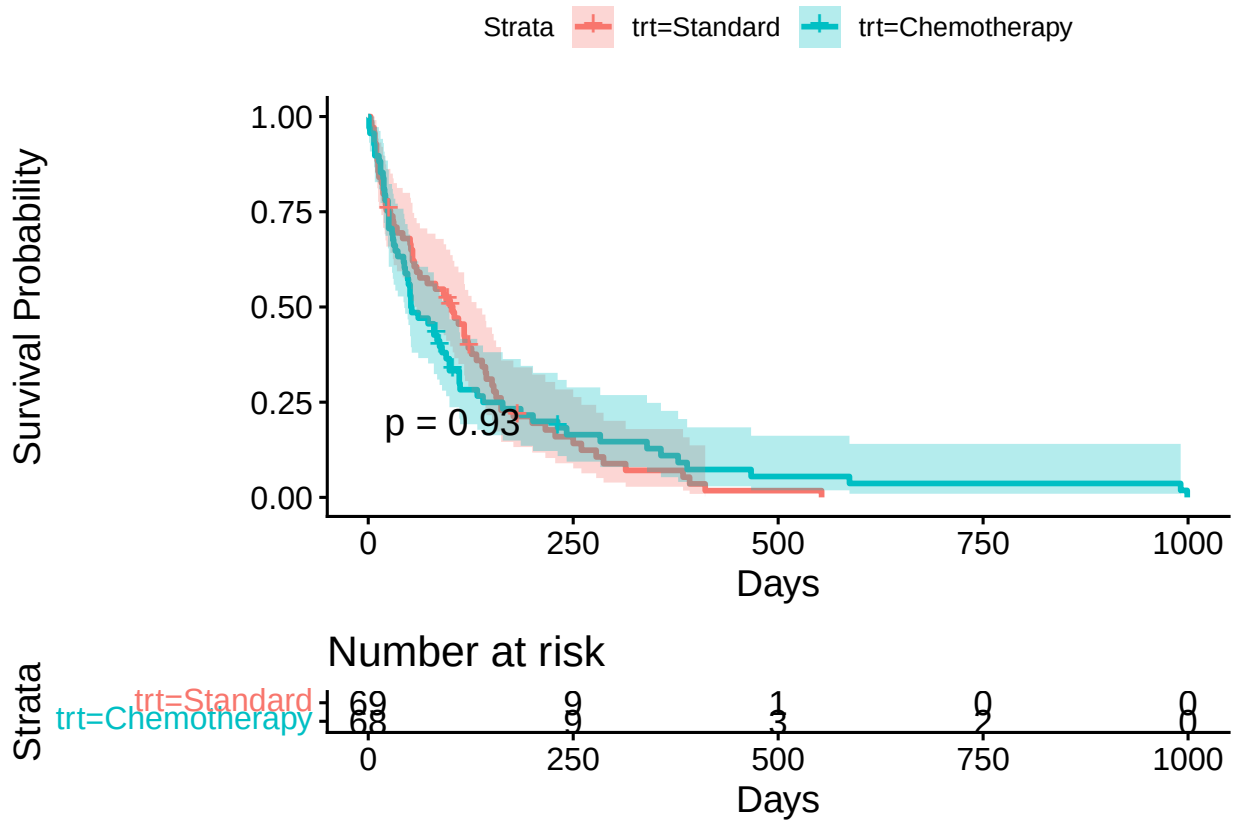
```
rmst_365
```

I have no Idea what the object 'NOTE' is that isn't found within the data set, and I've looked all over the internet to find different ways to write this code. I also don't know what I could have done to have my arguments return Infinity on both the maximum and minimum.

C. Kaplan-Meier plot by treatment

```
km_trt <- survfit(
  Surv(time, status == "Dead") ~ trt,
  data = mutate(veteran_clean, trt = fct_recode(trt, Chemotherapy = "Test"))
)
ggsurvplot(
  km_trt,
  data = vctb,
  conf.int = TRUE,
  pval = TRUE,
  risk.table = TRUE,
  xlab = "Days",
  ylab = "Survival Probability"
)
```

```
## Ignoring unknown labels:
## * colour : "Strata"
```



Within these plots, we can see an illustration of the previous Kaplan Meier test, and the different survival rates between standard treatment and chemotherapy treatment with patients.